

Balanced Glass Design: A flavor perception changing system by controlling the center-of-gravity

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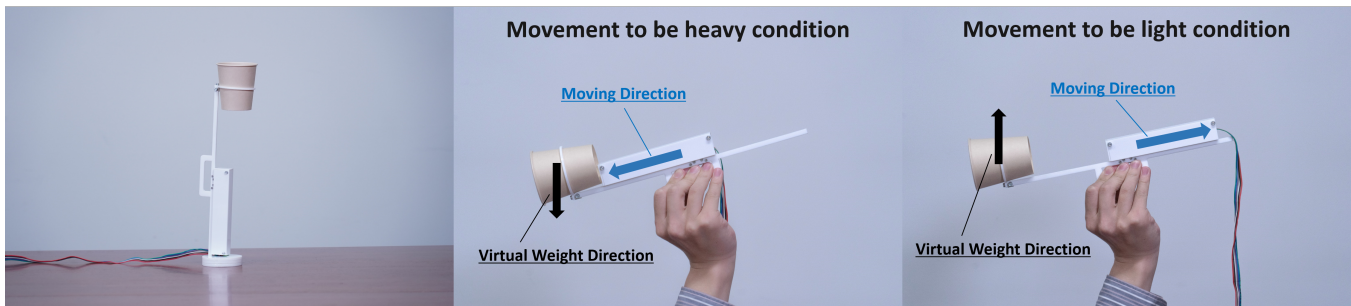


Figure 1: (Left) Appearance of Device. (Center) Mechanism for adding heaviness. (Right) Mechanism for adding lightness.

ABSTRACT

In this paper, we propose Balanced Glass Design, a system to change flavor perception. The system consists of glass-type device shifting its center of gravity in response to the user's motion which allows drinking a beverage with a virtual perception of weight through drinking motion. We thought It's possible to intervene in the user's perception of flavor by displaying virtual weight perception, and so conducted experiments on weight perception and demonstrations as a user study. This paper describes the system design, the result of experiments, and comments obtained through a user study.

CCS CONCEPTS

• Human-centered computing → Human computer interaction (HCI).

KEYWORDS

Cross-modal, Human Food Interaction, Virtual Reality

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1 INTRODUCTION

Flavor is composed of multi-modal perceptions. When eating or drinking something, we perceive the flavor through not only taste but also a combination of various perceptions. For example, we perceive spicy food by smell rather than taste. Another example is how we feel a potato chips spoiling their crispiness due to moisture, it's not tasty. From these examples, we can see that food perception is related to the senses of touch and smell.

In this study, we propose a system to change flavor perception through drinking a beverage with a perception of a virtual weight. The system consists of glass-type device. The device shifts its center of gravity causes a perceptual change in the beverage's weight and changes the flavor. Whether the perception of flavor can be changed by the perception of weight, it's based on the hypothesis of the cross-modal phenomenon in taste and tactile sensation. As a basis for this hypothesis, Harrar et al. conducted a study in which they examined the changes in taste when the size, weight, shape, and color of cutlery were changed. In this work, they found that the characteristics of the cutlery changed the thickness and saltiness of the yogurt, and also made it seem more luxurious [Harrar and Spence 2013]. Similar to the cutlery in their research, glass is an interface between humans and food. Therefore, these evidences suggest that if we could interactively control the information received from the glass, we could change the flavors of the beverage.

As a previous project, our research team proposed a system that enables people to eat food while presenting its weight [Hirose et al. 2015]. In the previous system, we showed that increasing the weight of the food by changing the center of gravity of the tableware affects the food's flavor through psychophysical experiments. Namely, in our previous work, we focused on tableware and realized the change of food perception from weight perception. In this research, we have extended the previous method to beverages, and improved the method not only for heaviness but also for lightness.

In the demonstration, we focused on the perception of wine. The taste of wine is very complex, and there is a unique vocabulary used to express the flavors. For example, "heavy" or "light" are often used as a metaphor. However, no one has ever actually tasted a physically "heavy" or "light" wine. Through the experience of perceptual change, the concept of the work we want to offer is to understand that perceptions other than taste have a potential to change the sensation of a beverage.

2 RELATED WORK

In recent years, many studies have been reported that aim to influence the perception of food from factors other than taste. For example, Narumi et al. proposed an AR/VR system to improve satiety by changing the apparent size of cookies [Narumi et al. 2012]. This research group also proposed a perception change system to transform the appearance of cookies and scent with AR technology [Narumi et al. 2011]. An example of applying thermal sensations, Suzuki et al. proposed a system to improve the flavor richness and aftertaste strength of beverages by presenting thermal sensations to the skin [Suzuki et al. 2014]. As a study that focuses on how to eat, Vi et al. proposed a gustatory interface using acoustic levitation and examined changes in the perception of sweet, bitter, and umami tastes [Vi et al. 2017]. Lin et al. applied digital fabrication technology to create cookies with different internal structures to enhance satiety [Lin et al. 2020].

The above research case can be considered as a computerized intervention in flavor perceptions. When we perceive food or drink, we perceive it through the tableware, the structure of the food, the sense of smell, touch, and visual information, so there are ways to intervene in flavor perception by applying computational control. If we can successfully intervene in the perception, and if we can connect enhancement of other perceptions (ex. taste and satisfaction, enjoyment), it is possible that we can realize new food experiences.

In this study, we implemented a mechanism changing center-of-gravity and constructed a flavor perception change system. In recent years, there have been many reports on VR systems with the moment change based on changing the center-of-gravity [Fujinawa et al. 2017; Huang et al. 2020; Shigeyama et al. 2019; Zenner and Kruger 2017]. The same mechanism was also used in our previous research, and it has been confirmed that weight can be effectively presented [Hirose et al. 2015]. Since this kind of weight presentation method requires no strings or external forces, it can give weight without restricting the user's movement. Therefore, it is an effective method for VR controllers and interfaces that users touch directly. In this study, we implemented a system that can present a wide dynamic range of weights by implementing a mechanism in which the power motor itself moves, similar to the work of Shigeyama et al. [Shigeyama et al. 2019].

3 BALANCED GLASS DESIGN

In this section, we describe the mental models and the system details. The system consists of a glass-type device that can change the the center of gravity, allowing the user to drink while feeling its heaviness or lightness. When adding heaviness, the system shifts the center of gravity in the user's direction following the drinking motion (Figure 1. Center). In contrast, when presenting lightness,

the center of gravity is shifted toward the glass's bottom (Figure 1. Right). This weight shifting changes the moment of inertia at the grasping part's fulcrum, which results in changing weight corresponding to the weight or lightness of the tip of the module. The purpose of the device is to change the food perception by intervening in the parameter, heaviness or lightness of the drink, which it's actually hard to change.

3.1 Mental Models

This system aims to change the flavor perception, i.e., taste, satisfaction, and deliciousness, from a weight perception change. As to whether it is possible to induce a change in food perception from a change in weight perception, we hypothesized a mental model of the unconscious predictions for food and beverage. For example, when we eat a cream puff, we bite into it expecting a soft crunch, and at that moment, we don't think the cream puff is hard as a rock. Besides, If we were to eat the cream puff by grabbing it with our hands, we would expect to feel the lightness and thus predict the fluffiness crunch. In this series of actions, it is difficult to change the perception of the food itself. However, in the case of food prediction, it is possible to intervene by changing the center of gravity. This mental model aims to change the flavor perception through the illusion that "I drank it while thinking it was heavy," or the unexpectedness that "I expected it to be heavy, but found it to be light at the moment I drank it".

The system aims to change the food perception by the weight perception due to the center of gravity changing, which means that we are using the illusion twice. At first, the weight presentation changes only the moment of inertia balance, and the overall mass does not change. Second, the purpose of the weight presentation is to change the level of satisfaction, taste, etc., which is entirely different from weight perception. Considering the mental model, the addition of weight, for example, gives the illusion that the density of the food has increased. Therefore, the food may taste richer or more satisfying, as it did in our previous research [Hirose et al. 2015].

3.2 System Details

We describe the processing flow of our system. We used a Raspberry Pi 4 as the computer board. The accelerometer (MPU 9250) installed inside the glass-type device (figure 2) measures the tilt angle. According to the device's tilt, the motor slider (RSA0N11M9A0K) shifts and changes the moment. The motor slider's position was controlled with PID control of the current value through the motor driver (DR8833), and the position of the motor slider measured through an A/D converter IC (MCP3208).

In designing the center of gravity shift mechanism, the balance between the entire device's mass and the moving object is essential. If the overall mass is too large or the moving object is too light, users cannot feel the change in the moment. It is desirable to make the overall mass light and the moving part as heavy as the actuator's power will allow. To address this issue, we designed the entire device to be driven by the motor slider itself as a weight because the motor slider itself is the heaviest component in the whole device. Figure 2 shows the inside of the drive part. The weight of the entire device is 127[g], and the weight of the moving part to change the center

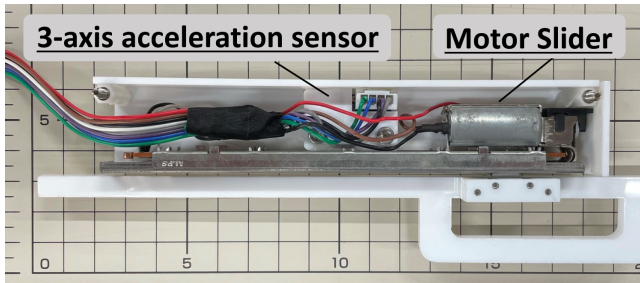


Figure 2: The inside of the device. The actuator is used as a weight for moment shift.



Figure 3: A participant in the experiment

of gravity is 100[g], and it moves 100[mm]. Through an extreme implementation in which 79[%] of the mass of the entire device is shifted, we were able to achieve a dynamic change in the center of gravity, even though the overall mass of the device is 127[g], which is about the same as that of an ordinary wine glass. To reduce the device's overall weight and achieve a hygienic demonstration, we adopted a disposable paper cup that can be attached to the part that touches the user's mouth (Figure 1).

4 EXPERIMENT

Herein, we describe a psychophysical experiment to clarify how much weight the device is capable of presenting. In the experiment, we adopted the adjustment method.

- (1) The subjects hold a device shifting the center of gravity in their left hand and other devices not moving (Slider Position is 0) in their right hand.
- (2) The subjects adjust the tip position's weight so that the counterbalance of both hands is the same.
- (3) Perform this operation for the center-of-gravity presentation position 0 to 10.

This result shows the relationship between the center of gravity of the module and the amount of weight that can be presented. Figure 3 is a subject taking part in an experimental study. Figure 4 is showing the position of the center of gravity. We show the result of this experiment in Figure 3.

As a result of the experiment, we found that the moment's weight presentation increased steadily from 0 to 4 and 9 to 10 positions. On the other hand, the subject could not feel any moment change between 4 and 9. For the 4 to 9, it may be because the moving object is close to the handle from the 4 to 9 position. In this position, even if it moves, the change in the moment is slight. The results show that the device can present weight of 24.4[g] by position changes from 0 to 4, and present 31.7[g] for the entire device. Considering the weight of a sip of the drink, we can assume that it will give enough weight.

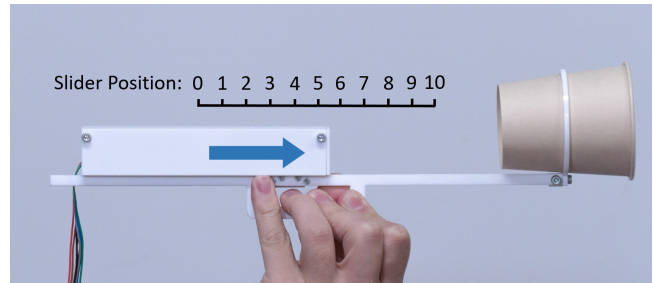


Figure 4: The slider position in the experiment

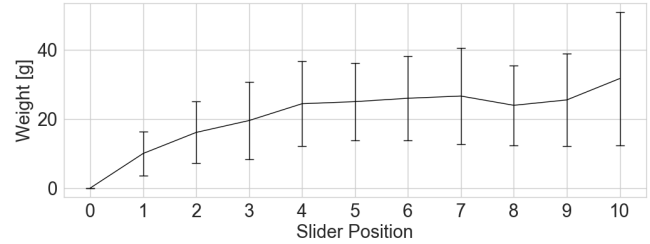


Figure 5: Experimental Results. Relationship between the moment and the presented weight. Maximum presented weight is 31.7[g]

5 DEMONSTRATION

We demonstrated the system to 20 people. The participants chose five kinds of beverages (tea, juice, sports drink, coffee, and wine) and drank them in two different ways: (1) Adding heaviness or (2) Adding lightness. In the case of drinking while presenting heaviness or lightness, in advance, the participants drank the beverage in a state where the center of gravity shift is not, because we intend to make it easier to understand the change in flavor. We show the participants of the demonstration in Figure 6. The following are some of the comments we obtained through the demonstration.

Comment for Adding Heaviness

- Since I felt weight was added while I was drinking it, I felt as if the amount of drink was not decreasing. I felt like I drank a lot. (Tea)
- I felt like the drink was rushing towards me . It felt similar to the feeling of drinking from a plastic bottle or tumbler. I reflexively hesitated to tilt the glass because I felt like I might spill the drink. (Tea)
- I felt that the wine was more dry. This may have been because the changing weight distribution made me drink the wine with a different speed and volume than usual. (Wine)
- I couldn't perceive the weight, but I felt like the taste got stronger. It tasted better. (Juice)
- The speed at which the liquid entered my mouth changed relative to without the device and this caused the taste to change. It felt strange. I felt that it tasted better. (Tea)
- I felt like I was drinking something viscous. I feel like the cup's capacity is changing. (Sports Drink)
- When tipping the empty module, it feels like there's water in the cup even when there isn't.

Comment for Adding Lightness

- *I couldn't perceive a sense of lightness in terms of weight, but I felt like the taste was lighter. (Juice)*
- *It certainly felt lighter, but it just felt like I was drinking from just a light glass. (Tea)*
- *When I lighten it, I feel like the amount of drink does not decrease. (Tea)*
- *It slowed my drinking speed and moderated how the drink spread in my mouth. (Tea)*

Overall

- *I felt a clear change in the coffee and wine. I think this is because coffee and wine have a complex taste. On the other hand, it was difficult to feel a change with sports drinks and juices because their flavor is simple. (Wine, coffee, Sports Drink, Juice)*
- *The way the drink touched my mouth changed, so it obviously changed the taste as well. (Tea)*
- *If we could make the motor slide more smoothly and precisely according to the drinking motion, we could present weight to the user without them noticing. I thought, if such control were possible, it would be possible to cause behavioral changes or changes in flavor without the user noticing. (Tea)*

Most of the participants were able to perceive changes in heaviness and lightness. In addition, there were clear flavor perception changes from the weight perception. As several people pointed out, there was an apparent change in the way the glass was tilted due to the weight change when drinking, and thus the taste seemed to change as well. Some feedback showed that the effect of perceptual change differed depending on the drink.

6 CONCLUSION

In this paper, we proposed Balanced Glass Design, a system to change a beverage's flavor by presenting virtual weight perception. We described the system design, the mental model, the experiments on weight perception, and the comments from the system demonstration participants. The results of the demonstration suggested the possibility of the flavor perception change. In the future, we plan to have more participants experience this system to clarify the mental model's general characteristics. With the results, we would like to conduct psychophysical experiments on the mental models' features and scientifically verify the flavor perception change and mental models.

REFERENCES

- Eisuke Fujinawa, Shigeo Yoshida, Yuki Koyama, Takuji Narumi, Tomohiro Tanikawa, and Michitaka Hirose. 2017. Computational Design of Hand-Held VR Controllers Using Haptic Shape Illusion. In *Proceedings of the 23rd ACM Symposium on Virtual Reality Software and Technology (Gothenburg, Sweden) (VRST '17)*. Association for Computing Machinery, New York, NY, USA, Article 28, 10 pages. <https://doi.org/10.1145/3139131.3139160>
- Vanessa Harrar and Charles Spence. 2013. The taste of cutlery: how the taste of food is affected by the weight, size, shape, and colour of the cutlery used to eat it. *Flavour* 2, 1 (June 2013), 1–13 pages. <https://doi.org/10.1186/2044-7248-2-21>
- Masaharu Hirose, Karin Iwazaki, Kozue Nojiri, Minato Takeda, Yuta Sugiura, and Masahiko Inami. 2015. Gravitamine Spice: A System That Changes the Perception of Eating through Virtual Weight Sensation. In *Proceedings of the 6th Augmented Human International Conference (Singapore, Singapore) (AH '15)*. Association for Computing Machinery, New York, NY, USA, 33–40. <https://doi.org/10.1145/2735711.2735795>



Figure 6: Participants in the demonstration

- Kai-Chieh Huang, Chen-Kuo Sun, Da-Yuan Huang, Yu-Chun Chen, Ruei-Che Chang, Shuo-wen Hsu, Chih-Yun Yang, and Bing-Yu Chen. 2020. Glissade: Generating Balance Shifting Feedback to Facilitate Auxiliary Digital Pen Input. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376505>
- Ying-Ju Lin, Parinya Punpongsonan, Xin Wen, Daisuke Iwai, Kosuke Sato, Marianna Obrist, and Stefanie Mueller. 2020. FoodFab: Creating Food Perception Illusions Using Food 3D Printing. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (Honolulu, HI, USA) (CHI '20)*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376421>
- Takuji Narumi, Yuki Ban, Takashi Kajinami, Tomohiro Tanikawa, and Michitaka Hirose. 2012. Augmented Perception of Satiety: Controlling Food Consumption by Changing Apparent Size of Food with Augmented Reality. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (Austin, Texas, USA) (CHI '12)*. Association for Computing Machinery, New York, NY, USA, 109–118. <https://doi.org/10.1145/2207676.2207693>
- Takuji Narumi, Shinya Nishizaka, Takashi Kajinami, Tomohiro Tanikawa, and Michitaka Hirose. 2011. Meta Cookie+: An Illusion-Based Gustatory Display. In *Virtual and Mixed Reality - New Trends*, Randall Shumaker (Ed.). Springer Berlin Heidelberg, Berlin, Heidelberg, 260–269.
- Jotaro Shigeyama, Takeru Hashimoto, Shigeo Yoshida, Takuji Narumi, Tomohiro Tanikawa, and Michitaka Hirose. 2019. Transcalibur: A Weight Shifting Virtual Reality Controller for 2D Shape Rendering Based on Computational Perception Model. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems (Glasgow, Scotland UK) (CHI '19)*. Association for Computing Machinery, New York, NY, USA, 1–11. <https://doi.org/10.1145/3290605.3300241>
- Chie Suzuki, Takuji Narumi, Tomohiro Tanikawa, and Michitaka Hirose. 2014. Affecting Tumbler: Affecting Our Flavor Perception with Thermal Feedback. In *Proceedings of the 11th Conference on Advances in Computer Entertainment Technology (Funchal, Portugal) (ACE '14)*. Association for Computing Machinery, New York, NY, USA, Article 19, 10 pages. <https://doi.org/10.1145/2663806.2663825>
- Chi Thanh Vi, Asier Marzo, Damien Ablart, Gianluca Memoli, Sriram Subramanian, Bruce Drinkwater, and Marianna Obrist. 2017. TastyFloats: A Contactless Food Delivery System. In *Proceedings of the 2017 ACM International Conference on Interactive Surfaces and Spaces (Brighton, United Kingdom) (ISS '17)*. Association for Computing Machinery, New York, NY, USA, 161–170. <https://doi.org/10.1145/3132272.3134123>
- Andre Zenner and Antonio Kruger. 2017. Shift: A Weight-Shifting Dynamic Passive Haptic Proxy to Enhance Object Perception in Virtual Reality. *IEEE Transactions on Visualization and Computer Graphics* 23, 4 (April 2017), 1285–1294. <https://doi.org/10.1109/TVCG.2017.2656978>